Setting Up Jupyter Notebooks

In this document, we will walk you through setting up the infrastructure that you will need to complete the programming assignments and to follow through the labs. There are two options that we will cover: Anaconda (recommended option) and Google Colaboratory (Google Colab).

Anaconda is a free and open-source distribution of Python and R. Google Colab is a free cloud service. Both support running Jupyter notebooks, which is what we will be using throughout this course. If you choose the Anaconda option, the notebooks will be executed locally; Google Colab notebooks will be run on Google's cloud servers.

Anaconda (suggested option)

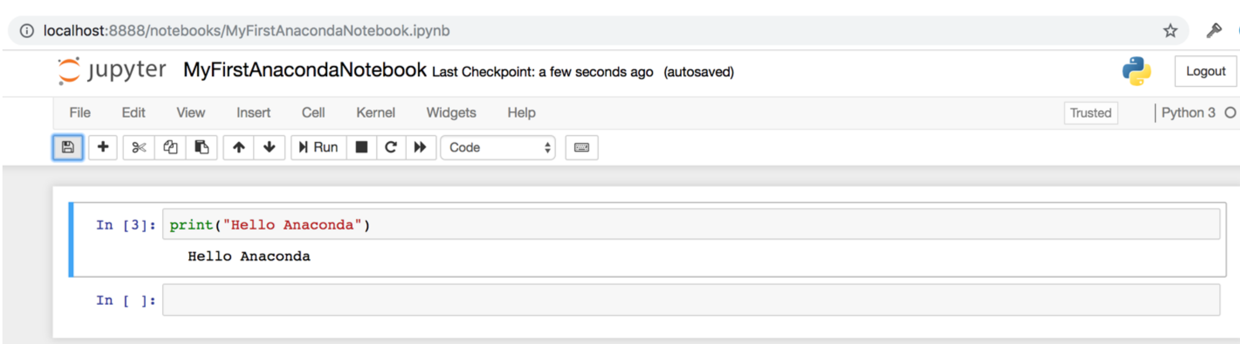
1. Follow the installation instructions at https://docs.anaconda.com/anaconda/install/

2. Once you have installed Anaconda, open the Anaconda Navigator and launch Jupyter Notebook. This will launch a new browser window (or a new tab) showing the Notebook Dashboard.

3. On the top of the right-hand side, there is a dropdown menu labeled "New → Python3". Create a new Notebook with the Python version you installed.

4. Rename your Notebook. Either click on the current name and edit it or find rename under File in the top menu bar. You can name it to whatever you'd like, but for this example we'll use MyFirstAnacondaNotebook.

5. In the first line of the Notebook (also call cell), type or copy/paste `print("Hello Anaconda")`. To run the code inside this cell, click on the cell and press Shift+Enter.



6. Save your Notebook by either clicking the save and checkpoint icon or select File → Save and Checkpoint in the top menu.

7. You can run a whole program by clicking the Run button or selecting Cell → Run All (from the top menu).

8. **To close a Jupyter Notebook**: from Jupyter Notebooks top menu bar, select File → Close and Halt.

9. **To upload an existing Jupyter Notebook**: Click on upload in the Notebook Dashboard and choose the .ipynb file. You can also go to File → Open and choose the .ipynb file. For example, load the notebook for the first recitation, named "Rec1_Python_2020.ipynb".

10. To open the loaded Jupyter Notebook, you can click on the name of the .ipynb file in the Notebook Dashboard (once the file is uploaded).

11. **To close the Notebook Dashboard**: Click the Quit button at the upper right of the Notebook Dashboard and close the window or tab.

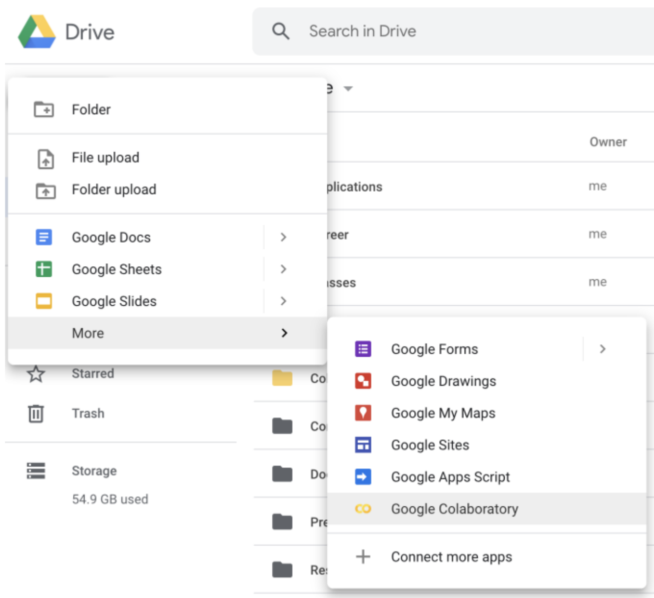
12. **To close the Navigator**: from Navigator's top menu bar, select Anaconda Navigator - Quit Anaconda-Navigator.

Google Colab

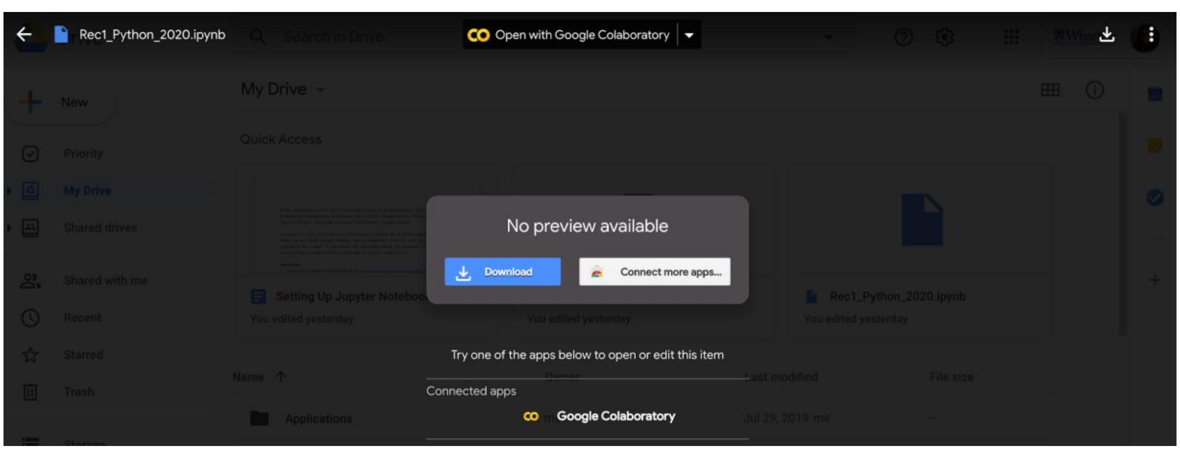
Although we strongly suggest you use Anaconda to run Python code, you could also use Google Colab if you prefer not to use your computer's processing resources and run your code in the cloud. You can do so using Google Colab following the instructions below.

You will need a Google account.

1. Go to your Google Drive
2. To create a new notebook, click on New → More → Google Colaboratory.



3. Alternatively, you could upload an existing .ipynb file to your Google Drive and choose 'Open with Google Colaboratory.'



4. You should be able to execute code in the cells.

5. To ensure that you will be able to access the data files that you'll be using for your analysis, you will need to mount your drive. Say for instance, your data is uploaded to your drive in a folder called Data, you could use the following code snippet:

   from google.colab import drive

   import os

   drive.mount('/content/Drive', force_remount=True)

   os.chdir('/content/Drive/My Drive/Data')

   You will be prompted to enter an authorization code. If this is succesful, you will be able to load the data from your Google Drive.

Before mounting your drive, you will not be able to access the file 'College.csv'

```
[3] data = pd.read_csv('College.csv')
```

```
---------------------------------------------------------------------------
FileNotFoundError                         Traceback (most recent call last)
<ipython-input-3-cff6ec5d79c9> in <module>()
----> 1 data = pd.read_csv('College.csv')

                              4 frames
/usr/local/lib/python3.6/dist-packages/pandas/io/parsers.py in __init__(self, src, **kwds)
   1889         kwds["usecols"] = self.usecols
   1890
-> 1891         self._reader = parsers.TextReader(src, **kwds)
   1892         self.unnamed_cols = self._reader.unnamed_cols
   1893

pandas/_libs/parsers.pyx in pandas._libs.parsers.TextReader.__cinit__()

pandas/_libs/parsers.pyx in pandas._libs.parsers.TextReader._setup_parser_source()

FileNotFoundError: [Errno 2] File College.csv does not exist: 'College.csv'
```

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After mounting your drive, you will be able to access the file 'College.csv'

```
[4] # When you run this block, you will need to click open a link to get an authentication code
    from google.colab import drive
    drive.mount('/content/Drive', force_remount=True)

    # Say for instance, your data is uploaded to a folder named Data
    import os
    os.chdir('/content/Drive/My Drive/Data')

    data = pd.read_csv('College.csv')
```

```
Mounted at /content/Drive
```